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Lab Assignment 6 - 2ip Nand & 2:1 Mux using NAND gate

Objective:

- The objective of this lab is to make more progress in cadence virtuoso. Within this lab we are tasked with creating a schematic of a 2 input NAND and a 2:1 Mux using 3, 2 input NAND gates. We are to simulate and verify the results of the inputs and outputs while also observing the difference between the two schematics.

Procedures:

1.0 - Create a Schematic

- Using Cadence Virtuoso we will create a schematic using pmos, nmos, input and output pins.
- For a 2 input NAND we will place two pMOS in parallel and 2 nMOS in series.
- For a 2:1 mux we will need 3 input pins (A,B, and Select), which will be driven into 3 NAND gates. Specifically the Select input is followed by an inverter that will then use the complement and non complement as inputs. We drive each of the first two NANDS with one input and a select or select'. The final NAND takes the output of the two previous NANDS.

2.0 - Create a Testbench Schematic

- Within this test bench schematic we will use the symbol generator to apply our sources to the schematic which includes: vdd, gnd, inputs and outputs. With each input we will apply a voltage pulse to allow for multiple tests when generating a waveform. As well as setting each input with a period delay to have multiple variations when testing for proper output. The vdd and gnd will be constant as vdd is 5v and gnd is 0v. We then select all our inputs and outputs to be measured.

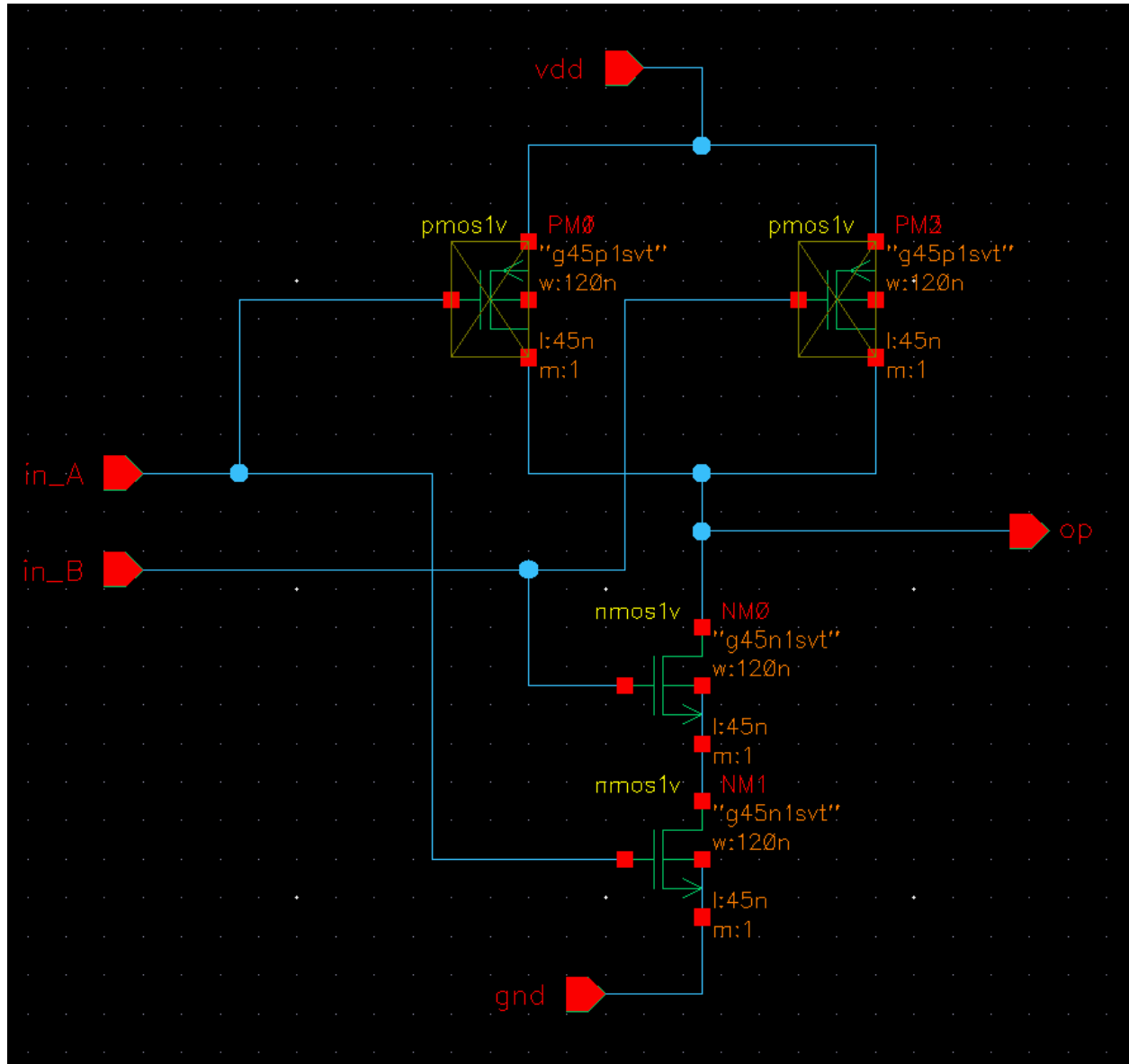
3.0 - Analyze Waveform for Correctness

- Once a waveform has been generated. To verify a correct waveform the number of waves should correspond equally to the number of inputs and outputs. Since in both of these there are differing inputs and 1 output, there should be a total of 4 waves to compare for the 2:1 mux and only 3 waves for the 2 input NAND.

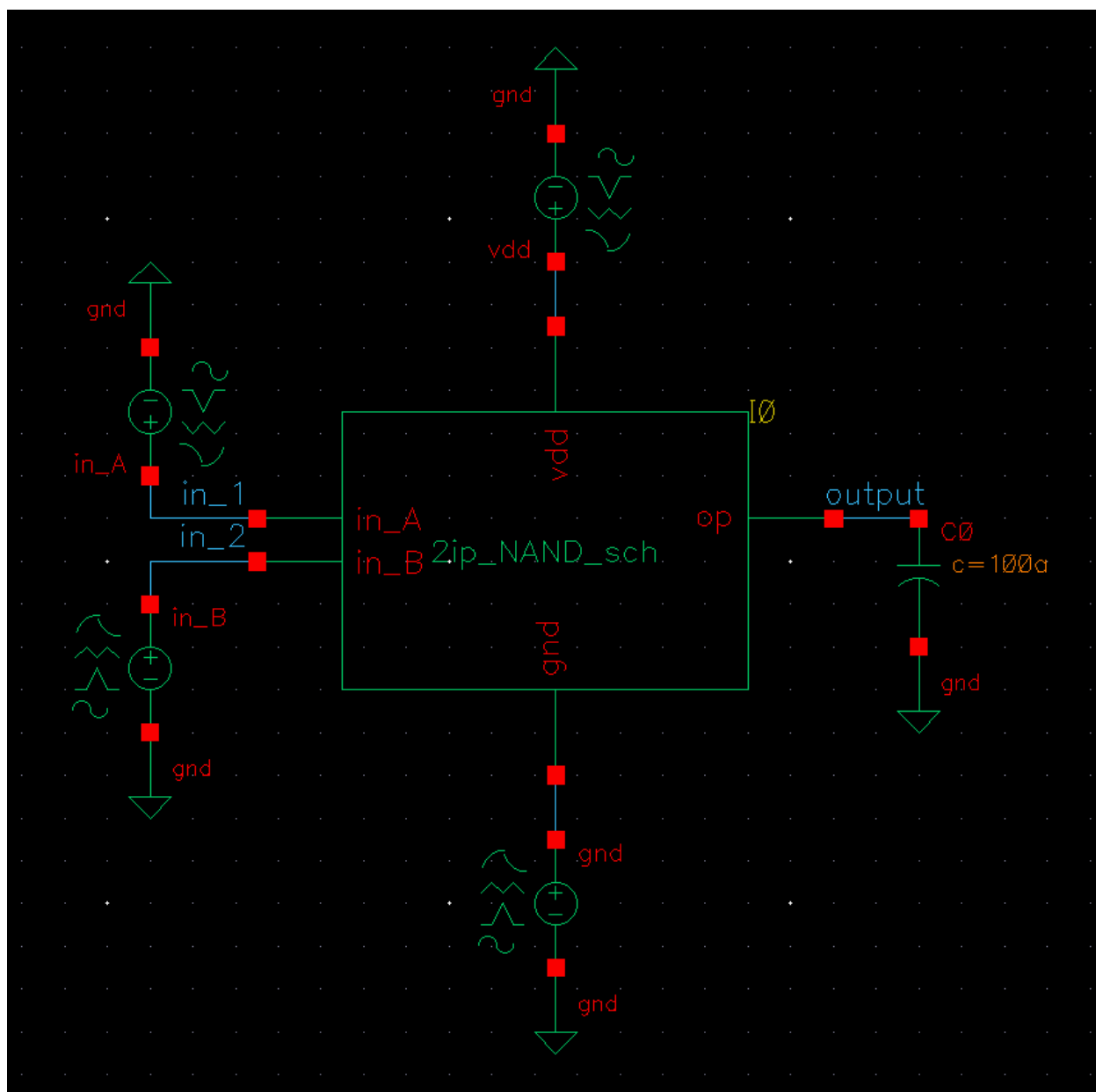
Design description:

Total 2:1 Mux Transistors: 7 PMOS + 7 NMOS = 14 Transistors

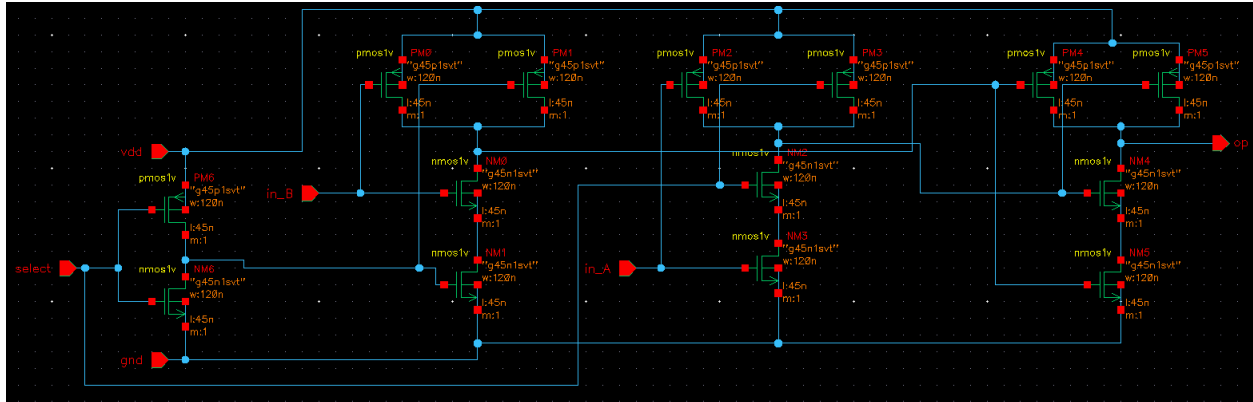
Total 2 input NAND Transistors: 2 PMOS + 2 NMOS = 4 Transistors



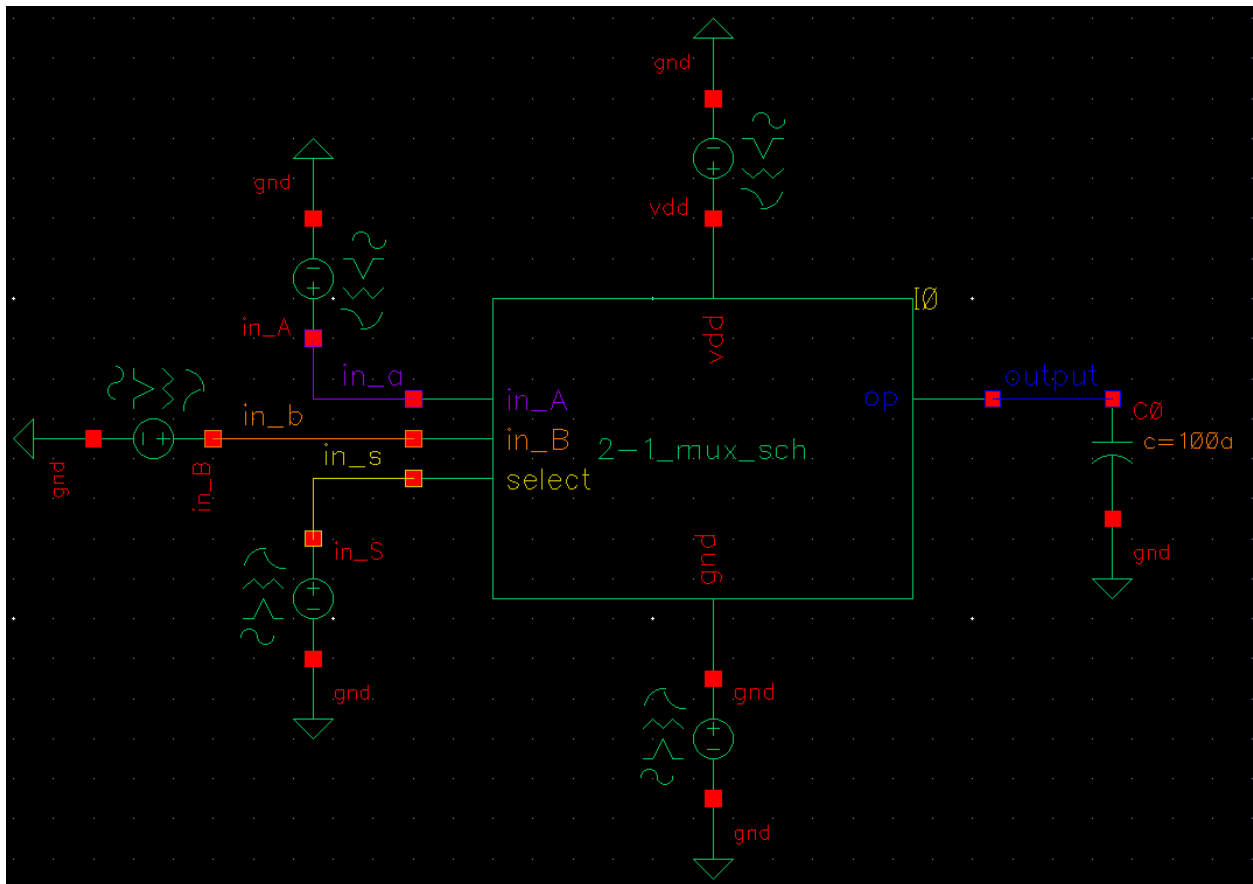
2-Input NAND Schematic



2-Input NAND Test Circuit / Symbol

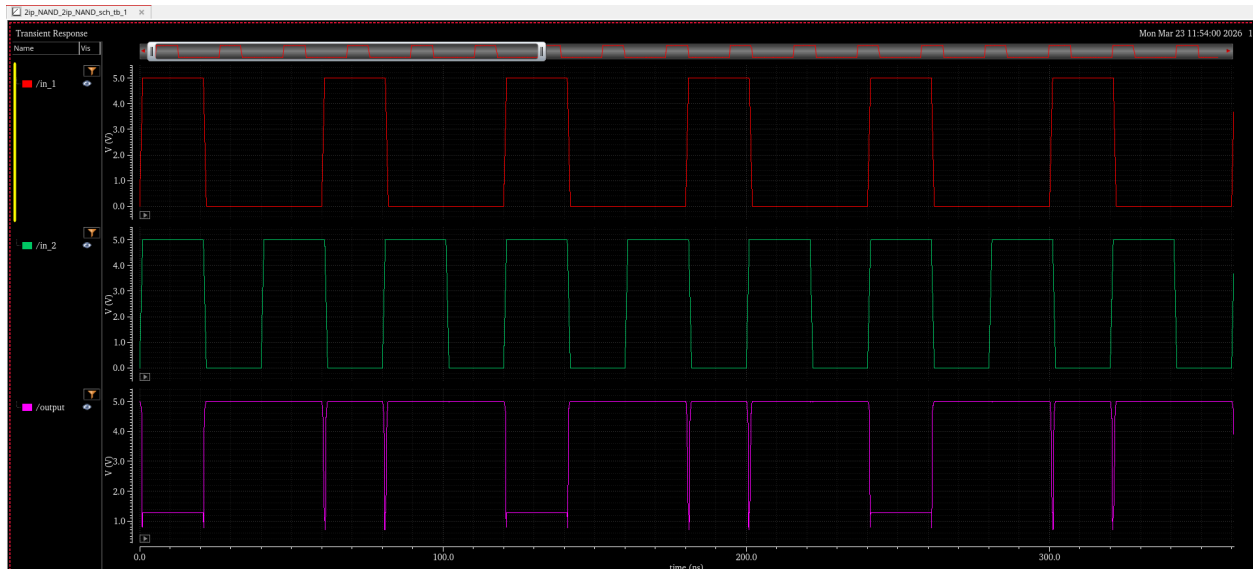


2:1 Mux Schematic



2:1 Mux Test Circuit / Symbol

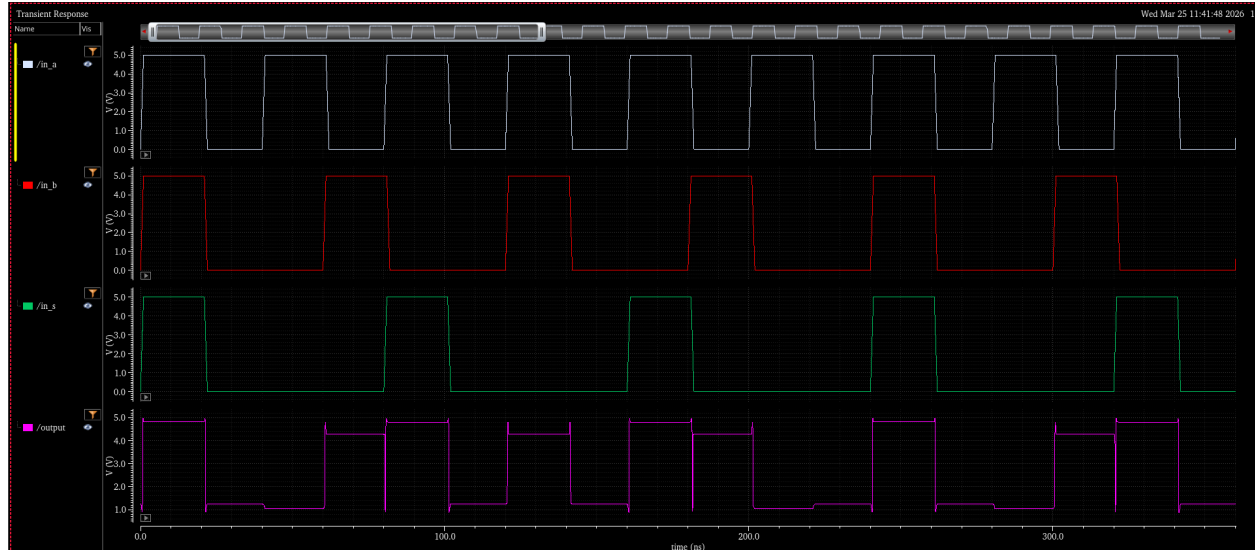
Results & Observations:



2 Input NAND Waveform

Observation:

- Looking at the waveform for the 2 input NAND we can see that we have three waveforms. Waveforms are labeled with our inputs A and B, while the final waveform is our output. When we analyze the graph we can see that the behavior of our inputs changes the output based on the transistors. When both A and B are high (1); then, we can see that the output is low (0) because of the nmos being on and pointing toward ground. When our inputs have any A or B set to low (0), the pMOS is on and a path to vdd is directed toward the output. Some spikes in the graph are due to transitioning not being aligned causing a short until aligned properly.
- Waveform Period for Inputs: A: 80ns, B: 60ns, C: 40ns



2:1 Mux Waveform

Observation:

- The 2:1 mux is an interesting graph as there is a lot happening. When we look at the structure of our waveforms we have 3 inputs (A, B, and Select) with one output wave. S is the select and when select is low (0) then input B is followed for the output. When select is high (1), then input A is followed for the output. In the graph we can see the changes in the output when the S waveform changes. In the beginning input S shows high so output shows input A which is also high, next input S goes low for a long delay and output reflects B, B starts out low then jumps high and the output follows that same logic until S changes back to high to follow A again.

Conclusion:

- This lab was another way to get familiar with cadence virtuoso but with more complex circuits. We were tasked with making a simple two input NAND gate and verifying it through simulation. Then we were tasked with making two input cascading NAND to make a mux. This 2:1 mux was a complex circuit as we needed to be careful with where we put our inputs and where we put our outputs as then became inputs for another NAND. I learned that with simple gates and cascading we can make more complex circuits and simulate the results to verify they were done correctly. We reinforced our CMOS logic design, signal propagation, and the comparison from schematic to real circuit behavior.